

$^2\text{H}(^{34}\text{S},\text{p}\gamma)$ 1973Wa10

$^{34}\text{S}(\text{d},\text{p})^{35}\text{S}$ in inverse kinematics.

1973Wa10: A 59.6-MeV ^{34}S beam was produced from the BNL MP-tandem Van de Graaff facility. Targets were $200\text{ }\mu\text{g}/\text{cm}^2$ TiD prepared by evaporating titanium onto the Cu, Al, and Mg target backings in a deuterium atmosphere. γ rays were detected using a 35 cm^3 Ge(Li) detector at 0° with FWHM=2 keV at 656 keV. Measured E_γ . Deduced $T_{1/2}$ for two ^{35}S levels using Doppler Shift Attenuation lineshape analysis.

 ^{35}S Levels

<u>$E(\text{level})$</u>	<u>J^π[†]</u>	<u>$T_{1/2}$[‡]</u>	<u>Comments</u>
0	$3/2^+$		
1574	$1/2^+$	2.29 ps 35	$T_{1/2}$: Lifetime=3.3 ps 5 from 1973Wa10.
2350	$3/2^-$	0.90 ps 14	$T_{1/2}$: Lifetime=1.3 ps 2 from 1973Wa10.

[†] From the Adopted Levels.

[‡] Using Doppler Shift Attenuation Method (DSAM).

 $\gamma(^{35}\text{S})$

<u>E_γ</u>	<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>E_f</u>	<u>J_f^π</u>
776	2350	$3/2^-$	1574	$1/2^+$
1574	1574	$1/2^+$	0	$3/2^+$

 $^2\text{H}(^{34}\text{S},\text{p}\gamma)$ 1973Wa10Level Scheme